

Discrete sensory inputs can stimulate both simple and complex behaviors

A **behavior** is an action carried out by muscles under control of the nervous system. Examples include an animal using its throat muscles to produce a song or releasing a scent to mark its territory. Behavior is an essential part of acquiring nutrients and finding a partner for sexual reproduction. Behavior also contributes to homeostasis, as when honeybees huddle to conserve heat (see Concept 40.3). In short, all of animal physiology contributes to behavior, and behavior influences all of physiology.

Many behaviors, especially those involved in recognition and communication, rely on specialized body structures or form. For instance, the throat pouch of a male frigatebird (see Figure 51.1) is essential for courtship. Inflating the pouch exposes its vibrantly colored surface and provides a resonating chamber that amplifies the male's drumming. As this example illustrates, the process of natural selection that shapes behaviors also influences the evolution of animal anatomy.

What approach do biologists use to determine how behaviors arise and what functions they serve? The Dutch scientist Niko Tinbergen, a pioneer in the study of animal behavior, suggested that understanding any behavior requires answering four questions, which were highlighted in Figure 51.1:

1. What stimulus triggers the behavior and how do the various body systems bring it about?
2. How does the animal's experience during growth and development influence the response to the stimulus?
3. How does the behavior aid survival and reproduction?
4. What is the behavior's evolutionary history?

The first two questions ask about *proximate causation*—how a behavior occurs or is modified. The last two questions ask about *ultimate causation*—why a behavior occurs in the context of natural selection.

Experiments on proximate causation by Tinbergen earned him a share of a Nobel Prize awarded in 1973. Such studies defined the field of *ethology*, the study of animal behavior observed in a natural environment. We'll consider those and

related experiments in the early part of the chapter. The concept of ultimate causation is central to **behavioral ecology**, the study of the ecological and evolutionary basis for animal behavior. We'll explore this vibrant area of modern biological research in the rest of the chapter.

Fixed Action Patterns

In addressing Tinbergen's first question, the nature of the stimuli that trigger behavior, we'll begin with behavioral responses to well-defined stimuli, starting with an example from one of Tinbergen's own experiments.

As part of his research, Tinbergen kept fish tanks containing three-spined sticklebacks (*Gasterosteus aculeatus*), a species in which males, but not females, have red bellies. Male sticklebacks attack other males that invade their nesting territories (**Figure 51.2**). Tinbergen noticed that his male sticklebacks also behaved aggressively when a red truck passed within view of their tank. Inspired by this chance observation, he carried out experiments showing that the red color of an intruder's underside is the proximate cause of the attack behavior. A male stickleback will not attack a fish lacking red coloration, but will attack even unrealistic models if they contain areas of red color. The territorial response of male sticklebacks is an example of a **fixed action pattern**, a sequence of unlearned acts directly linked to a simple stimulus. The fixed action pattern of certain moths takes the form of evasive flight maneuvers—loops and spirals—performed instantaneously upon hearing the sounds of an echolocating bat or an ultrasonic whistle in the same frequency range. Fixed action patterns are essentially unchangeable and, once initiated, are usually carried to completion. The trigger for the behavior is an external cue called a **sign stimulus**, such as a red object that prompts a male stickleback's aggressive behavior.

Migration

Environmental stimuli not only trigger behaviors but also provide cues that animals use to carry out those behaviors. For example, a wide variety of birds, fishes, and other animals use environmental cues to guide **migration**—a regular, long-distance change in location. In the course of migration, many animals pass through environments they have not

► **Figure 51.2 Sign stimuli in a classic fixed action pattern.** (a) A male stickleback fish attacks other male sticklebacks that invade its nesting territory. (b) Models, whether realistic or not, can trigger the aggressive behavior, but only if the underside is red. Researchers concluded that the red belly of the intruding male is a sign stimulus that releases the aggressive behavior.

? Suggest an explanation for why this behavior might have evolved (its ultimate causation).

